

B.Tech. III-Sem. Main/Back/Reback Exam(UD) 2023**Branch-CSE****Subject Code: CSL204****Subject Name: Microprocessor & Microcontroller****Maximum Marks: 100****Time: 3 Hours***Instructions to candidates: -*

PART A: Short answer questions (up to 25 words) 10 x 2 marks = 20 marks. All ten questions are compulsory.

PART B: Analytical/Problem Solving questions (up to 100 words) 6 x 5 marks = 30 marks. Candidates have to answer six questions out of eight.

PART C: Descriptive/ Analytical/Problem solving questions 5 x 10 marks = 50 marks. Candidates have to answer five questions out of seven.

The following code(s) are required:

1. _____ 2. _____

Part- A**(Answer should be given upto 25 words only)****[10×2=20]****All questions are compulsory**

- Q1. Explain the role of DMA controller with neat diagram?
- Q2. What is the role of accumulator in microprocessor?
- Q3. Define general purpose registers.
- Q4. Why AD₀ and AD₇ lines are multiplexed?
- Q5. What is the use of ALE signal?
- Q6. Define Instruction and instruction set.
- Q7. List some technical features of 8085 microprocessor?
- Q8. What do you mean by input port and output port?
- Q9. What is the role of Instruction Register in 8085 microprocessor?
- Q10. What is the role of stack pointer (SP) and program counter (PC)?

Part-B

(Analytical/Problem solving questions)

[6×5=30]

Attempt any six questions

- Q1. Draw and explain the pin diagram of 8085 μ P.
Q2. Explain various flags available in 8085 μ P.
Q3. Explain the different types of buses in 8085 μ P with neat diagram.
Q4. What is Subroutine? Explain the use of stack in CALL and RETURN Instructions in detail.
Q5. Differentiate between Microcontroller and Microprocessor.
Q6. Explain various addressing modes in 8085 μ P with suitable examples.
Q7. Explain the following opcodes with suitable examples:
a) LDA b) ADD c) ADC d) SHLD e) MVI
Q8. How will you demultiplex the address and data bus in 8085 μ P?
How can you interface 1024 Bytes RAM with 8085 μ P?

Part-C

(Descriptive/Analytical/Problem solving/Design questions)

[5×10=50]

Attempt any five questions

- Q1. Explain the architecture of 8251 (USART) and explain the components. <https://www.rtuonline.com>
Q2. Draw the architecture diagram of 8085 μ P and explain the components.
Q3. Explain interrupts of 8085 μ P in detail.
Q4. Write an assembly language program to add two 8-bit numbers with carry.
Q5. Explain 8255 programmable peripheral interface in detail.
Q6. Draw the timing diagram of the instruction MVI A, 32H.
Q7. Write short note on any 2 of the following:
a) Memory in 8085 μ P
b) RIM and SIM instructions in 8085 μ P
c) Counters and Time Delays